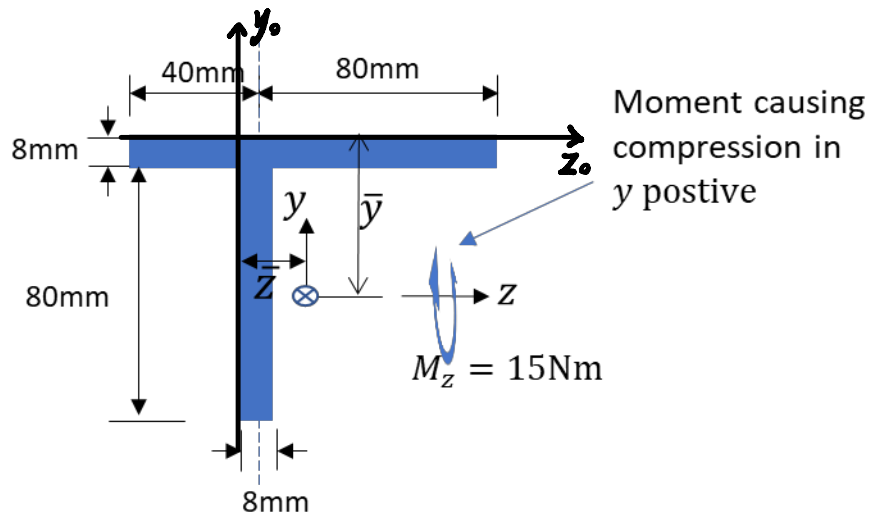


Structures I

Tutorial Sheet 5: Axial Stress in Beams

1. Calculate the maximum stress for the section below.



$$\sigma_x = - \bar{M}_z y$$

$$\bar{y} = \frac{4(8 \cdot 120) + 48(8 \cdot 80)}{(8 \cdot 120) + (8 \cdot 80)} = 21.6 \text{ (mm)}$$

$$\bar{z} = \frac{24(8 \cdot 120) + 4(8 \cdot 80)}{(8 \cdot 120) + (8 \cdot 80)} = 13.6 \text{ (mm)}$$

$$\therefore I_z = \left[\frac{(120 \text{ mm})(8 \text{ mm})^3}{12} + (4 \text{ mm} - 21.6 \text{ mm})^2 (8 \text{ mm} \cdot 120 \text{ mm}) \right]$$

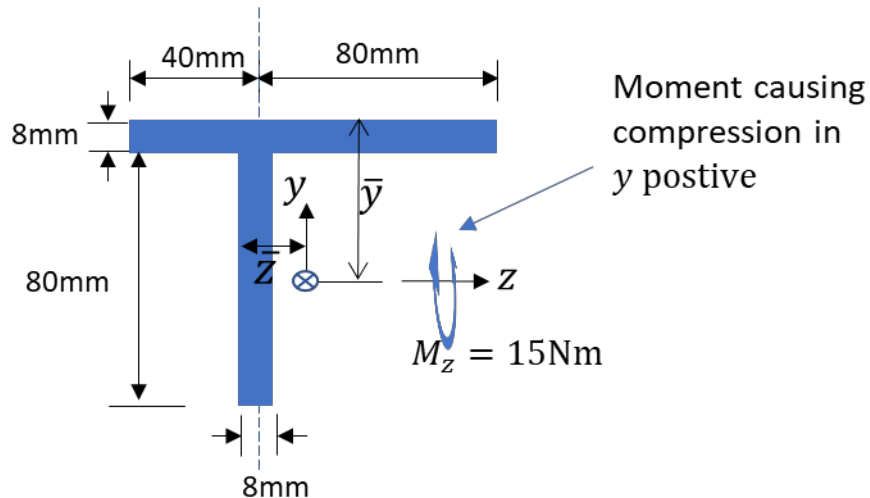
$$+ \left[\frac{(8 \text{ mm})(80 \text{ mm})^3}{12} + (48 \text{ mm} - 21.6 \text{ mm})^2 (8 \text{ mm} \cdot 80 \text{ mm}) \right]$$

$$= 1.127 \times 10^{-6} \text{ m}^4$$

Structures I

Tutorial Sheet 5: Axial Stress in Beams

1. Calculate the maximum stress for the section below.



$$\begin{aligned}
 \therefore I_y &= \left[\frac{(120 \text{ mm})^3 (8 \text{ mm})}{12} + (24 \text{ mm} - 13.6 \text{ mm})^2 (8 \text{ mm} \cdot 120 \text{ mm}) \right] \\
 &\quad + \left[\frac{(8 \text{ mm})^3 (80 \text{ mm})}{12} + (4 \text{ mm} - 13.6 \text{ mm})^2 (8 \text{ mm} \cdot 80 \text{ mm}) \right] \\
 &= 1.31 \times 10^{-6} \text{ m}^4
 \end{aligned}$$

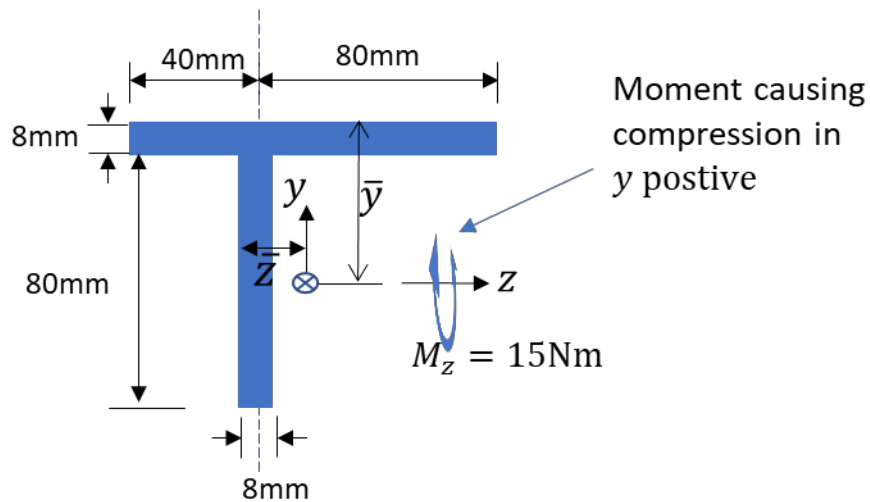
$$\begin{aligned}
 \therefore I_{zy} &= 0 + (8 \text{ mm} \cdot 120 \text{ mm}) (4 \text{ mm} - 21.6 \text{ mm}) (20 \text{ mm} - 13.6 \text{ mm}) \\
 &\quad + (8 \text{ mm} \cdot 80 \text{ mm}) (-48 \text{ mm} + 21.6 \text{ mm}) (4 \text{ mm} - 13.6 \text{ mm}) \\
 &= -2.703 \times 10^{-7} \text{ m}^4
 \end{aligned}$$

$$\therefore \bar{\sigma}_z = \frac{I_y M_x - I_{zy} M_y}{I_x I_y - I_{zy}^2} = 1.404 \times 10^7 \text{ N/m}^2$$

Structures I

Tutorial Sheet 5: Axial Stress in Beams

1. Calculate the maximum stress for the section below.

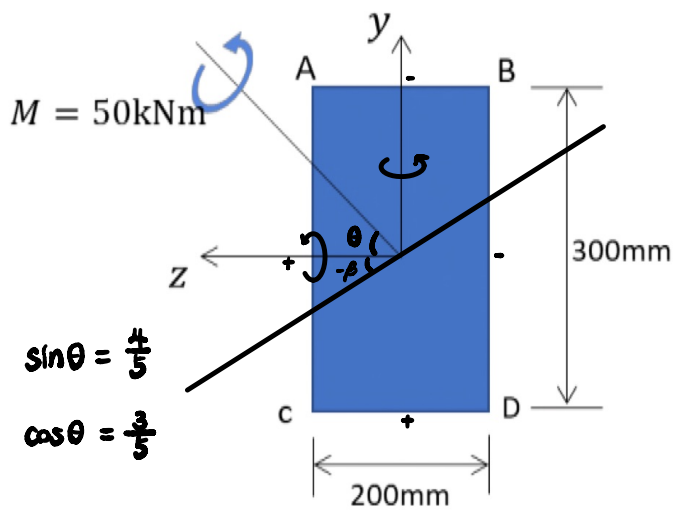


$$\sigma_x = -1.404 \times 10^7 y$$

$$\therefore \sigma_{x \max} \text{ when } y = 21.6 \text{ mm} - 88 \text{ mm} = -66.4 \text{ mm}$$

$$\therefore \sigma_{x \max} = 9.6 \times 10^5 \text{ N/m}^2$$

2. Find the bending stress at the four corners of the rectangular section beam shown in figure below. Also, calculate the inclination of the neutral axis with respect to the centroidal axis.



$$I_y = \frac{(300 \text{ mm})(200 \text{ mm})^3}{12} = 2 \times 10^{-4} \text{ m}^4$$

$$I_z = \frac{(200 \text{ mm})(300 \text{ mm})^3}{12} = 4.5 \times 10^{-4} \text{ m}^4$$

$$I_{yz} = 0 \text{ m}^4$$

$$M_y = M \sin \theta = 40 \text{ kNm}$$

$$M_z = M \cos \theta = 30 \text{ kNm}$$

$$\therefore \bar{M}_y = \frac{I_z M_y - I_{yz} M_z}{I_z I_y - I_{yz}^2} = 2.00 \times 10^7 \text{ N/m}^2$$

$$\bar{M}_z = \frac{I_y M_z - I_{yz} M_y}{I_z I_y - I_{yz}^2} = 6.67 \times 10^7 \text{ N/m}^2$$

$$\therefore \beta = \tan^{-1} \left(-\frac{\bar{M}_y}{\bar{M}_z} \right) = -16.70^\circ$$

$$\therefore \sigma = \bar{M}_y z - \bar{M}_z y$$

$$\therefore \sigma_A = \cancel{8 \times 10^6 \text{ N/m}^2} \quad 10 \text{ MPa}$$

$$\sigma_B = \cancel{-1.2 \times 10^7 \text{ N/m}^2} \quad -30 \text{ MPa}$$

$$\sigma_C = \cancel{1.2 \times 10^7 \text{ N/m}^2} \quad 30 \text{ MPa}$$

$$\sigma_D = \cancel{-8 \times 10^6 \text{ N/m}^2} \quad -10 \text{ MPa}$$